

Chia-Yu Hsu

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Education

Ph.D. in University of Michigan - Ann Arbor, Advisor: Shubhanshu Shekhar

Michigan, USA

Department of Electrical Engineering and Computer Science

2025 - Present

- GPA: 3.93/4.0 (2 semesters so far)
- Research Topic: Advances in anytime-valid testing, estimation, and learning

M.S. in National Taiwan University (NTU), Advisor: I-Hsiang Wang

Taipei, Taiwan

Graduate Institute of Communication Engineering

2019-2021

- Overall GPA: 3.81 / 4.3
- Research Topic: Information theory and statistics

Work Experience

Research Assistant

Hsinchu, Taiwan

National Yang Ming Chiao Tung University (NYCU), *Reinforcement Learning and Bandits Lab*

2024 - 2025

- Studied global convergence of stochastic policy optimization in reinforcement learning, with emphasis on optimization and probabilistic analysis of learning algorithms.

Research Assistant

Taipei, Taiwan

National Taiwan University, *Networked Information and Communication Lab*

2023 - 2024

- Developed theory for active sequential hypothesis testing, focusing on adaptive data-source selection, statistical reliability, and sample-efficient decision making.
- Characterized optimal tradeoffs among error exponents under heterogeneous data sources; work appeared in IEEE Transactions on Information Theory and ISIT.

Wi-Fi System Designer

Hsinchu, Taiwan

Realtek Semiconductor Corp., *Wi-Fi Cross-Layer Team*

2021 - 2023

- Redesigned an IEEE 802.11ax spatial-reuse system and developed new algorithms that achieved up to **1.4× throughput improvement**; contributed to two patent submissions and feature adoption in a new chip design.
- Developed a power-saving rate-adaptation algorithm that reduced PHY-layer transmission power by more than **30%**; contributed to one patent submission
- Worked on cross-layer wireless system design, algorithm development, performance evaluation, and optimization for production-oriented Wi-Fi platforms.

Professional skills

Machine learning, Statistics and large language models, Reinforcement learning, Generative AI (under studying in Fall 26 semester), Wireless communication and system design, Python, SQL (under studying in Fall 26 semester)

Selected Research and Publications

- **Efficient Sequential Evaluation of Large Language Models** (2026 preprint): developed a sequential framework for evaluating LLMs with adaptive question selection and anytime-valid statistical guarantee.
- **Classifier-Based Nonparametric Sequential Hypothesis Testing** (2026 preprint): developed a nonparametric sequential hypothesis testing framework using a trained classifier. The framework can be applied to quickest change detection and anomaly detection.
- **Active Nonparametric Two-Sample Testing by Betting on Heterogeneous Data Sources**, IEEE ISIT 2026, oral presentation.
- **Tradeoffs among Action Taking Policies Matter in Active Sequential Multi-hypothesis Testing: the Optimal Error Exponent Region**, IEEE Transactions on Information Theory, 2025.

Selected Honors

ISIT Student Travel Grant (2026), Best Paper Award, NST Taiwan (2024), Best Undergraduate Project Award, NTUST Taiwan (2019), College Student Research Scholarship, NTUST Taiwan (2018)